

Pixelwise Split Conformal Prediction for Global Temperature Emulation via Leave-One-Out Ensemble Distillation

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Abstract

Neural emulators of climate model output rarely come with statistically principled uncertainty estimates. We study how *split conformal prediction* (SCP) can be repurposed from a post-hoc uncertainty wrapper into a *training-time spatial prior* for ensemble distillation. Given nine CMIP6 Earth System Models, we train nine leave-one-out (LOO) UNet++ teachers and calibrate each on its held-out target member, producing per-pixel calibration quantiles with the standard marginal split-CP validity interpretation at each spatial location and for each teacher, under exchangeability. From these calibration residual statistics, we derive calibration-informed center weights that score, for each pixel, how consistently each teacher tracks the bias-corrected ensemble consensus, and pixel-loss weights that reweight the distillation objective so that the student is pushed harder on historically difficult regions. On global monthly near-surface air temperature emulation across a 192×288 grid, the student achieves a test MAE of 0.9625 K versus 0.9782 K for a strong UNet++ baseline (−1.61%), decreases the error of 96.4% of grid cells, and reaches empirical pixelwise coverage of 90.6%. This student coverage is reported as an empirical calibration check rather than as a formal conformal guarantee for the distilled predictor. Our source code is available at <https://github.com/bunyaminkorkut/COPA-Global-Temperature-Emulation>.

Keywords: conformal prediction, uncertainty quantification, climate emulation, knowledge distillation, UNet++, ensemble learning, CMIP6

1. Introduction

Global climate projections underpin policy decisions on adaptation, mitigation, and risk assessment (Rolnick et al., 2022). Physics-based Earth System Models (ESMs) that constitute the Coupled Model Intercomparison Project Phase 6 (CMIP6; Eyring et al., 2016) are computationally expensive to run; data-driven emulators trained on their output can reproduce large-scale temperature fields at a fraction of the cost (Rasp and Thuerey, 2021; Schneider et al., 2017). However, most neural emulators produce only *point predictions*, with no principled statement of uncertainty. This is a critical gap when model output informs climate risk analyses that explicitly require probability distributions or interval forecasts (Gneiting and Katzfuss, 2014).

CMIP6 ensembles offer a natural avenue for uncertainty estimation: the nine ESMs used here differ in their physical parameterisations and therefore sample different plausible climate trajectories. A standard approach trains one neural network on all ensemble members jointly (Unal et al., 2023), but this collapses inter-model spread into a single point estimate. An alternative is to maintain the full ensemble and derive prediction intervals from the spread (Lakshminarayanan et al., 2017), yet ensemble-based intervals are not guaranteed to achieve a pre-specified coverage level for finite samples and arbitrary input distributions.

Split conformal prediction (SCP; Papadopoulos et al., 2002; Vovk et al., 2005) addresses this limitation: given a calibration set that is exchangeable with future test points, SCP wraps any point predictor with an interval whose marginal coverage is at least $1 - \alpha$ without distributional assumptions. Applying SCP to spatial climate outputs raises two design questions. First, the calibration data used to compute conformity scores should not be used as target supervision for the model being calibrated. Second, applying a single global calibration threshold wastes the strong spatial heterogeneity of emulator errors: tropical oceans are comparatively easy to emulate, while Arctic sea-ice regions, the Southern Ocean, and high-altitude terrain are systematically harder.

This paper addresses both issues. Our central methodological idea is to **repurpose split conformal calibration as a training-time spatial prior**: instead of using SCP only to produce intervals at test time, we extract spatial signals from the calibration phase and feed them into ensemble distillation. Concretely, our contributions are:

1. **A leave-one-out ensemble for temperature forecasting that provides a leakage-controlled basis for split conformal calibration.** We train nine UNet++ (Zhou et al., 2018) teachers from nine CMIP6 Earth System Models. Each teacher excludes one ESM from the target supervision set and is calibrated on that excluded target member. This design avoids direct training–calibration target leakage, while exchangeability remains approximate because of temporal autocorrelation and inter-model distribution shift (Section 5.1).
2. **Pixelwise split conformal calibration** yielding per-pixel quantile maps $Q_i(h, w)$ for each LOO teacher (Section 5.2). We frame the resulting validity statement as a per-pixel, per-teacher *marginal* guarantee under exchangeability, not as joint spatial coverage and not as conditional coverage.
3. **Empirical validation** on global emulation (Section 6): the distilled student improves over a strong UNet++ baseline (Unal et al., 2023) in all five latitudinal bands and

on 96.4% of grid cells, with empirical pixelwise coverage of 90.6% when evaluated with teacher-calibration-derived intervals. This coverage is interpreted as an empirical calibration check, not as a formal guarantee for the aggregated or distilled predictor.

2. Background: Split Conformal Prediction

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a dataset of input-output pairs drawn i.i.d. or, more generally, exchangeably from some distribution $P_{X,Y}$. Split conformal prediction (Papadopoulos et al., 2002; Vovk et al., 2005; Angelopoulos and Bates, 2023) partitions $\mathcal{D}$ into a proper-training set $\mathcal{D}_{\text{tr}}$ and a calibration set $\mathcal{D}_{\text{cal}}$ of size n_{cal} . A base model $\hat{f}$ is fitted on $\mathcal{D}_{\text{tr}}$. For each calibration example $(x_i, y_i) \in \mathcal{D}_{\text{cal}}$, the nonconformity score is

$$s_i = |\hat{f}(x_i) - y_i|. \quad (1)$$

Given a desired miscoverage level $\alpha \in (0, 1)$, the finite-sample-corrected quantile level is

$$q_{\text{level}} = \min\left(\frac{\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil}{n_{\text{cal}}}, 1\right), \quad (2)$$

and the calibration threshold is $\hat{q} = \text{Quantile}_{q_{\text{level}}}(\{s_i\}_{i=1}^{n_{\text{cal}}})$. For a new test input x_{n+1} , the conformal prediction interval is

$$C(x_{n+1}) = [\hat{f}(x_{n+1}) - \hat{q}, \hat{f}(x_{n+1}) + \hat{q}]. \quad (3)$$

The key guarantee (Vovk et al., 2005) is

$$\mathbb{P}\{y_{n+1} \in C(x_{n+1})\} \geq 1 - \alpha, \quad (4)$$

provided the calibration and test examples are exchangeable. No parametric model for $P_{X,Y}$ is required.

In our setting, x is a multi-channel spatio-temporal tensor and y is a two-dimensional temperature field. We apply SCP in a *pixelwise* manner by computing a separate calibration quantile $Q(h, w)$ for each spatial location (h, w) on the global 192×288 grid. Using absolute residuals as nonconformity scores yields symmetric prediction intervals whose widths adapt to local emulation difficulty, rather than relying on a single global threshold.

Scope of the validity claim. Throughout the paper, we claim only the *marginal* form of the SCP guarantee. Since calibration is performed independently at each spatial location, Equation (4) holds per pixel and per teacher, under exchangeability of the corresponding calibration and test nonconformity scores. We do *not* claim joint coverage of the full 192×288 field, conditional coverage at fixed inputs, or validity under distribution shift across ESMs. The distilled student requires an additional qualification. Because it is trained after calibration using teacher-derived calibration statistics, its reported 90.6% coverage should be interpreted as an empirical calibration check, not as a theorem inherited directly from split CP. In our LOO setup, exchangeability is therefore best viewed as an approximation: the construction removes direct target leakage between teacher training and calibration, but temporal autocorrelation and inter-ESM distribution shifts prevent the formal exchangeability condition from holding exactly.

3. Related Work

Split conformal prediction and distribution-free uncertainty. Split conformal prediction (Papadopoulos et al., 2002; Vovk et al., 2005; Angelopoulos and Bates, 2023) is a model-agnostic calibration framework that converts a point predictor and a held-out calibration set into prediction intervals with finite-sample marginal coverage under exchangeability. Our method uses the standard absolute-residual score for regression, but applies it independently at each spatial grid cell and for each LOO teacher. This keeps the validity statement transparent: the guarantee is per pixel and per teacher, not joint over the entire spatial field.

Conformal prediction for spatio-temporal weather fields. The closest methodological reference to our pixelwise calibration step is the conformal error-bar framework for neural weather models of Gopakumar et al. (2024). They calibrate neural weather forecasts over spatio-temporal output tensors in a point-wise manner, using absolute residual scores for deterministic models and applying the resulting quantiles as upper and lower error bars. Our pixelwise SCP step follows the same residual-quantile principle, but differs in two ways: we calibrate a leave-one-out ensemble of CMIP6 climate emulators rather than a single neural weather forecaster, and we reuse the resulting calibration residual statistics as training-time spatial signals for student distillation.

Locally adaptive and quantile-based conformal methods. Standard residual-based split CP gives symmetric intervals whose width is determined by calibration residual quantiles. This is simple and robust, but it does not explicitly learn asymmetric or input-conditional uncertainty. Conformalized quantile regression (CQR) (Romano et al., 2019) combines quantile regression with conformal calibration to obtain intervals that adapt to heteroscedasticity while retaining finite-sample marginal coverage. We do not train quantile regressors in the present work; instead, we use symmetric residual intervals because the target variable is near-surface air temperature. CQR is a natural extension for non-Gaussian or strongly heteroscedastic climate variables such as precipitation and wind extremes.

Conformal forecasting in meteorological applications. Conformal prediction has recently been used for uncertainty forecasting in meteorological settings. For example, Meng and Song (2024) apply conformal prediction to tropical cyclone track forecasting and compare multiple machine-learning models and conformal forecasting strategies. Their problem is trajectory-level uncertainty communication, whereas ours is grid-cell-level temperature emulation. Nevertheless, both settings illustrate the usefulness of conformal calibration as a lightweight alternative to expensive ensemble-based uncertainty quantification.

Temporal dependence and non-exchangeable conformal prediction. Climate fields are temporally dependent, so the exchangeability assumption of standard split CP is only approximate. Adaptive conformal inference (Gibbs and Candès, 2021), sequential predictive conformal inference (Xu and Xie, 2023), and more recent time-series conformal methods such as Transformer conformal prediction (Lee et al., 2024) explicitly address temporal dependence by updating or modelling residual quantiles over time. We do not use these methods in the present paper; instead, we make the approximate-exchangeability limitation explicit and leave sequential calibration for future work.

Conformal predictor ensembles and ensemble distillation. Conformal predictor ensembles have been studied through strategies such as cross-conformal prediction, bootstrap conformal prediction, and out-of-bag calibration (Linusson et al., 2020). Our ensemble is different: the LOO teachers are not combined to form a classical conformal predictor ensemble with pooled p-values or out-of-bag scores. Instead, each teacher is calibrated on its held-out target member, and the resulting calibration residual statistics are used to guide knowledge distillation. This connects conformal calibration with knowledge distillation (Hinton et al., 2015) and uncertainty-aware ensemble compression, including ensemble distribution distillation (Malinin et al., 2020), but retains a simpler residual-based conformal interpretation.

To the best of our knowledge, the combination of LOO target exclusion, pixelwise split conformal calibration, calibration-informed teacher weighting, and calibration-error pixel-loss weighting has not been studied together for global climate emulation. The novelty is therefore not the use of split CP alone, but the use of calibration residual statistics as a training-time spatial prior for distilling an ensemble of climate-model emulators into a single student.

4. Data and Model Setup

4.1. Dataset

We use monthly near-surface air temperature fields (`tas`) from nine CMIP6 ESMs: ACCESS-CM2, CNRM-CM6-1-HR, GISS-E2-1-H, MIROC-ES2L, NorESM2-MM, CESM2-WACCM, EC-Earth3-Veg, IPSL-CM6A-LR, and MPI-ESM1-2-HR, spanning the historical period 1850–2014. All fields are interpolated to a uniform 192×288 (latitude $\times$ longitude) grid at the native CESM2/CAM6 finite-volume resolution ($0.9375^\circ \times 1.25^\circ$).

For each target month, the dynamic input consists of nine ESM-specific lagged temperature channels constructed from the fixed lag set $\{t - 1, t - 12 \pm 2, t - 24 \pm 2, t - 36 \pm 2\}$, with one lagged temperature field provided per ESM, plus one static Digital Elevation Model (DEM) channel. The resulting tensor has 10 channels. The target is the temperature field at time t for the respective ESM. Evaluation is performed on months with index ≥ 1700 (approximately 1991–2014), yielding approximately $9 \times 280 = 2,520$ test samples.

4.2. Architecture

All models use the same UNet++ architecture (Zhou et al., 2018), with circular longitudinal padding to respect the periodic boundary condition of the globe. The model takes $10 \times 192 \times 288$ tensors as input and outputs a 192×288 temperature field. Training uses MSE loss with the Adam optimizer at a learning rate of 10^{-4} and weight decay 10^{-5} .

Following the model design and training protocol of (Unal et al., 2023), we train an original UNet++ baseline on all nine ESM target members simultaneously. Under this setup, the reproduced baseline reaches a best validation loss of 0.0540 and a test MAE of 0.9782 K.

5. Method

Figure 1 gives an overview of the full four-stage pipeline.

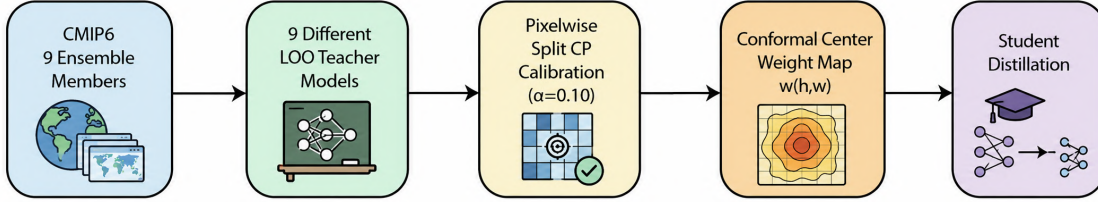


Figure 1: Pipeline overview. Nine LOO teachers are trained with one ESM excluded from target supervision, calibrated with pixelwise SCP on the excluded target member, combined through calibration-informed center weights, and used to distill a single student with pixel-loss weighting.

5.1. Stage 1: Leave-One-Out Teacher Training

The core difficulty in applying SCP to ensemble-trained models is that the calibration examples should not have been used as target supervision for the model being calibrated. We therefore train one teacher per ESM in a leave-one-out fashion. Teacher i (denoted f_i) is trained using the eight ESMs $\mathcal{E} \setminus \{e_i\}$ as target supervision and is calibrated exclusively on the excluded target ESM e_i . Thus, the excluded ESM is never used as an output target when training teacher f_i .

All nine ESM channels are nevertheless used as lagged covariates for every teacher. The LOO exclusion therefore applies to the *target supervision signal*, not to the input feature tensor. This design removes direct training–calibration target leakage and provides a practical basis for approximately treating calibration residuals as exchangeable with test residuals, while recognising that strict exchangeability is still limited by temporal autocorrelation and inter-model distribution shift.

Table 1 summarises the forecasting performances of nine LOO teachers.

The variation in calibration MAE reveals informational heterogeneity within the ensemble. LOO-2 (GISS-E2-1-H excluded as target) has the highest MAE (0.2006) and widest average conformal half-width (0.8353), indicating that GISS-E2-1-H is relatively difficult to emulate from the remaining target members. Conversely, LOO-3 (MIROC-ES2L excluded as target) has the lowest MAE (0.1682) and narrowest average half-width (0.6887). These differences are later reflected in the calibration-derived spatial diagnostics and pixel-loss weights.

5.2. Stage 2: Pixelwise Split Conformal Calibration

For each teacher f_i , we apply pixelwise SCP on its corresponding calibration set $\mathcal{D}_{\text{cal}}^{(i)}$ of size n_{cal} . This step is closest in spirit to point-wise conformal error bars for neural weather

Table 1: LOO teacher summary. Each model is calibrated on the ESM excluded from its target-supervision set. The conformal threshold targets nominal 90% marginal coverage under exchangeability. MAE, RMSE, and average half-width are reported in standardised-temperature units.

LOO	Excluded target ESM	MAE	RMSE	Avg. half-width
f_0	ACCESS-CM2	0.1846	0.2389	0.7644
f_1	CNRM-CM6-1-HR	0.1858	0.2405	0.7645
f_2	GISS-E2-1-H	0.2006	0.2607	0.8353
f_3	MIROC-ES2L	0.1682	0.2167	0.6887
f_4	NorESM2-MM	0.1762	0.2300	0.7427
f_5	CESM2-WACCM	0.1734	0.2257	0.7268
f_6	EC-Earth3-Veg	0.1860	0.2406	0.7711
f_7	IPSL-CM6A-LR	0.1947	0.2557	0.8176
f_8	MPI-ESM1-2-HR	0.1887	0.2447	0.7724

models (Gopakumar et al., 2024), where absolute residual scores are computed over spatio-temporal output tensors and the resulting quantiles are used as upper and lower prediction error bars. In our case, the output tensor is a monthly global `tas` field, and calibration is performed independently for each latitude–longitude grid cell and each LOO teacher. For each pixel (h, w) and time step t in the calibration set, the nonconformity score is

$$s_i(t, h, w) = |f_i(x_t)_{h,w} - y_t(h, w)|. \quad (5)$$

Applying Equation (2) with $\alpha = 0.10$, the pixel-level calibration quantile for teacher i is

$$Q_i(h, w) = \text{Quantile}_{q_{\text{level}}}(\{s_i(t, h, w)\}_{t=1}^{n_{\text{cal}}}), \quad (6)$$

yielding a 192×288 map of symmetric half-widths. At test time, the teacher-specific prediction interval at pixel (h, w) is

$$C_i(x; h, w) = [f_i(x)_{h,w} - Q_i(h, w), f_i(x)_{h,w} + Q_i(h, w)]. \quad (7)$$

Under exchangeability of the calibration and test scores at that pixel, the standard split-CP analysis applies independently at each (h, w) :

$$\mathbb{P}\{y_{n+1}(h, w) \in C_i(x_{n+1}; h, w)\} \geq 1 - \alpha.$$

This is a *per-pixel, per-teacher marginal* guarantee. It does not automatically imply joint coverage across pixels, conditional coverage at fixed climate states, or formal coverage for the distilled student.

Figure 2 summarises the spatial structure of the conformal interval diagnostics. The interval widths are largest at high latitudes and over complex orography, matching known regions of increased climate-model spread and emulator difficulty.

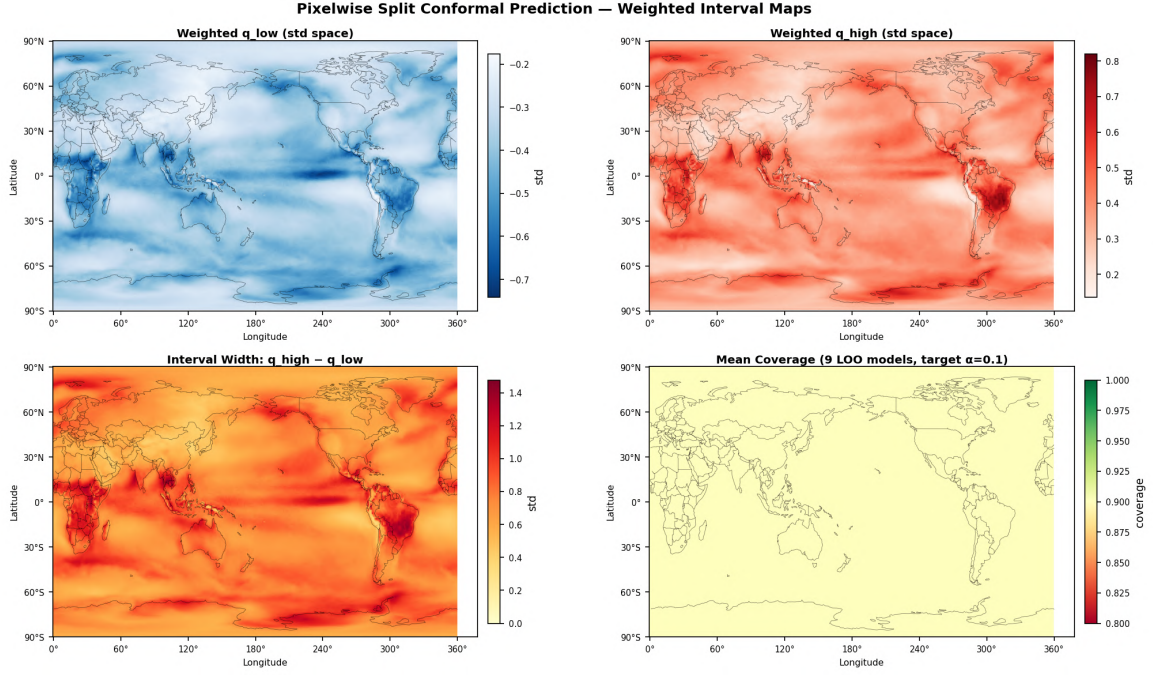


Figure 2: Pixelwise conformal interval diagnostics. The panels show the weighted lower interval endpoint, weighted upper interval endpoint, full interval width, and mean empirical coverage across LOO teachers. Larger interval widths indicate regions where calibration residuals require wider intervals to target 90% marginal coverage.

5.3. Stage 3: Calibration-Informed Center Weight Maps

The nine teachers can disagree substantially at a given pixel. We construct teacher combination targets by weighting bias-corrected teacher predictions according to their proximity to the ensemble consensus. This weighting is *calibration-informed* because the bias correction and spatial diagnostics are estimated on the LOO calibration residuals, but the dynamic softmax itself uses deviation from the bias-corrected teacher consensus.

Bias correction. Each teacher f_i may have a spatially varying mean bias. We estimate it over the calibration set and apply it as a pixel-level offset:

$$\delta_i(h, w) = \frac{1}{n_{\text{cal}}} \sum_{t=1}^{n_{\text{cal}}} (y_t(h, w) - f_i(x_t)_{h,w}), \quad \tilde{f}_i(x)_{h,w} = f_i(x)_{h,w} + \delta_i(h, w). \quad (8)$$

Dynamic ensemble weight. Let the bias-corrected teacher consensus be

$$\bar{f}(x)_{h,w} = \frac{1}{9} \sum_{i=1}^9 \tilde{f}_i(x)_{h,w}.$$

The instantaneous dynamic weight for teacher i is

$$w_i^{\text{dyn}}(t, h, w) = \frac{\exp\left(-\beta \left| \tilde{f}_i(x_t)_{h,w} - \bar{f}(x_t)_{h,w} \right| \right)}{\sum_j \exp\left(-\beta \left| \tilde{f}_j(x_t)_{h,w} - \bar{f}(x_t)_{h,w} \right| \right)}, \quad (9)$$

with temperature parameter $\beta = 2.0$. Teachers that deviate more strongly from the bias-corrected consensus are down-weighted exponentially.

Static diagnostic weight map. For inference-time diagnostics, we average the dynamic weights over all calibration time steps:

$$w_i^{\text{static}}(h, w) = \frac{1}{n_{\text{cal}}} \sum_{t=1}^{n_{\text{cal}}} w_i^{\text{dyn}}(t, h, w). \quad (10)$$

Figure 3 summarises the resulting spatial diagnostics, including dispersion in LOO weights, the pixel-loss map, and best/worst/mean teacher MAE maps. Figure 4 shows the dominant teacher index at each pixel.

5.4. Stage 4: Pixel-Loss Weighting and Student Distillation

Pixel-loss weights. To focus the student on high-error regions, we define a pixel-level loss weight proportional to the mean calibration MAE across all teachers:

$$p(h, w) = \frac{\frac{1}{9} \sum_{i=1}^9 \text{MAE}_i(h, w)}{\frac{1}{H \cdot W} \sum_{h,w} \frac{1}{9} \sum_{i=1}^9 \text{MAE}_i(h, w)}, \quad (11)$$

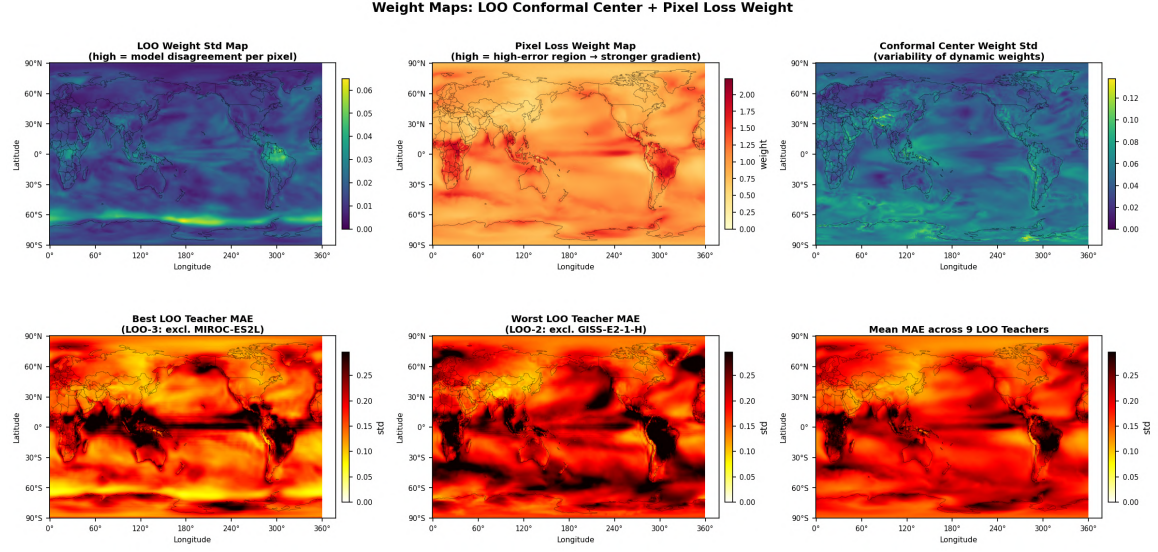


Figure 3: Calibration-derived spatial diagnostics. The panels summarise dispersion in LOO center weights, the pixel-loss weight map, dispersion in conformal-center weights, and the best, worst, and mean teacher MAE across the nine LOO teachers. Warmer colours in the MAE and pixel-loss panels indicate regions that receive stronger training emphasis during distillation.

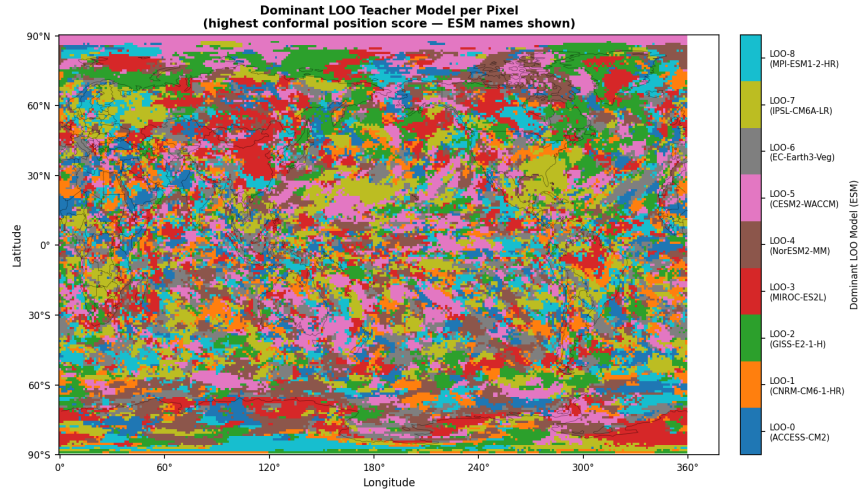


Figure 4: Pixelwise dominant teacher index, defined as the teacher i with the largest static calibration-informed center weight $w_i^{\text{static}}(h, w)$ at each pixel. Geographic clustering indicates regional variation in teacher behaviour.

where $\text{MAE}_i(h, w)$ is the mean absolute error of teacher f_i at pixel (h, w) over the calibration set. Normalising by the global mean preserves the total loss scale: polar and high-altitude pixels receive weights of roughly 2–3 $\times$ the global mean, while low-error tropical ocean regions are down-weighted.

Distillation objective. The student model g is trained to minimise

$$\mathcal{L} = \sum_{h,w} p(h, w) \left[(g(x)_{h,w} - y_{h,w})^2 + \lambda \left(g(x)_{h,w} - \sum_i w_i^{\text{dyn}}(t, h, w) \tilde{f}_i(x)_{h,w} \right)^2 \right], \quad (12)$$

where the first term is the supervised MSE loss against the ground truth y and the second term pulls the student toward the dynamically weighted teacher-consensus target. We use $\lambda = 1.0$.

The student is trained for 25 epochs with Adam (lr = 10^{-4} , weight decay 10^{-5} , batch size 8), achieving a best validation loss of 0.05212 compared with the baseline’s 0.05400.

6. Experiments

6.1. Evaluation Protocol

Following the evaluation protocol defined in (Unal et al., 2023), we evaluate all models on the CMIP6 test split, defined by months with index ≥ 1700 (approximately 1991–2014). The evaluation uses all nine ESM members, yielding approximately $9 \times 280 \approx 2,520$ test samples after lag construction. MAE and RMSE are reported after denormalising the predictions and targets back to Kelvin.

For empirical interval evaluation of the distilled student, we use the student prediction as the interval centre and the teacher-calibration-derived mean half-width as a static spatial uncertainty prior:

$$C_g(x; h, w) = [g(x)_{h,w} - \bar{Q}(h, w), g(x)_{h,w} + \bar{Q}(h, w)], \quad \bar{Q}(h, w) = \frac{1}{9} \sum_{i=1}^9 Q_i(h, w). \quad (13)$$

Because $\bar{Q}(h, w)$ is not calibrated directly on an independent calibration set for g , the resulting student coverage is empirical rather than a formal split-CP guarantee. Coverage and average half-width are computed over the full test set.

6.2. Global Results

Table 2 summarises the global test results. The student reduces global MAE by 1.61% relative to the baseline and improves predictions at 96.4% of grid cells. Its empirical pixelwise coverage is 90.6%, slightly above the nominal 90% target. As emphasised above, this is an empirical calibration check for the student interval in Equation (13), not a formal SCP guarantee for g .

Table 2: Global test results. MAE and RMSE are in Kelvin after denormalisation. Avg. half-width is the mean conformal half-width $\bar{Q}(h, w)$ in standardised-temperature units. Pixel-level Kelvin equivalents depend on the spatially varying denormalisation factor.

Model	MAE (K)	RMSE (K)	Improved pixels	Cov.	Avg. HW
Baseline (Unal et al., 2023)	0.9782	1.3048	—	—	—
Our method	0.9625	1.2854	96.4%	90.6%	0.7457

6.3. Regional Breakdown

Figure 5 and Table 3 break down MAE by latitudinal band. The student improves over the baseline in all five bands. The largest absolute gains are in the polar regions (-0.030 K in the Arctic and -0.022 K in Antarctica), where calibration residuals are larger and the pixel-loss weighting concentrates more training signal. Even the tropical band, where the baseline is already strongest, shows a small but consistent improvement of -0.008 K.

Table 3: Regional MAE comparison in Kelvin. Negative Δ means that the student has lower error than the baseline.

Region	Baseline (K)	Ours (K)	Δ (K)
Arctic ($> 70^\circ\text{N}$)	1.7957	1.7657	-0.0300
N. Hemisphere ($30\text{--}70^\circ\text{N}$)	0.5083	0.4981	-0.0100
Tropics ($30^\circ\text{S}\text{--}30^\circ\text{N}$)	0.2756	0.2680	-0.0080
S. Hemisphere ($30\text{--}70^\circ\text{S}$)	1.0343	1.0187	-0.0160
Antarctica ($< 70^\circ\text{S}$)	1.8057	1.7836	-0.0220

6.4. Spatial MAE Maps and Teacher Calibration

Figure 6 shows spatial MAE maps for the baseline and the student, their difference (baseline minus student), and the per-pixel MAE distribution. Green or blue improvement regions correspond to pixels where the student has lower error. Improvement is widespread and concentrated around high-gradient regions such as sea-ice margins, the Southern Ocean, and major orographic features.

Figure 7 summarises LOO teacher calibration. The panels report calibration MAE, RMSE, mean conformal half-width, and empirical coverage for each held-out target member. LOO-2 (GISS-E2-1-H) and LOO-7 (IPSL-CM6A-LR) have the widest intervals, indicating relatively larger calibration residuals for those held-out target members.

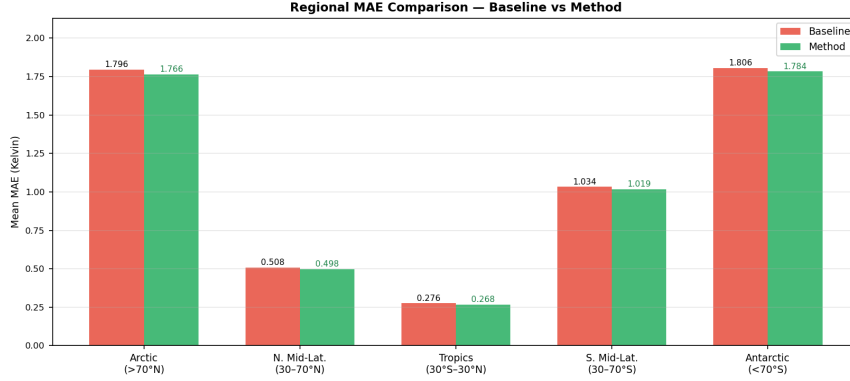


Figure 5: Regionalised MAE comparison by latitudinal band. The student improves over the baseline in all bands, with the largest absolute gains in polar regions.

6.5. Spatial Conformity Analysis

Figure 8 shows the zonal mean conformal half-width $Q(h, w)$ as a function of latitude, expressed in Kelvin after multiplying by the per-pixel climatological standard deviation. The profile reveals a clear spatial structure: polar regions are substantially more uncertain ($Q \approx 4.0\text{--}4.2\text{K}$ for Arctic and Antarctica) than the tropics ($Q \approx 0.9\text{K}$), consistent with the known difficulty of emulating sea-ice dynamics and high-latitude variability in CMIP6 models. The shaded band shows ± 1 standard deviation across longitudes at each latitude, indicating that within-band longitudinal variability is especially large in the Southern Ocean and along continental margins.

7. Limitations

We explicitly catalogue the limitations of the study so that the strength of the validity claim is not over-read.

Only marginal validity, only per pixel, only per teacher. The SCP guarantee invoked in Section 5.2 is the standard marginal one: it holds for each pixel and each LOO teacher separately, under exchangeability of calibration and test scores at that pixel. We do not prove joint coverage of the full 192×288 field, conditional coverage at a given climate state, or coverage under distribution shift across ESMs. Achieving these stronger forms remains challenging in conformal prediction (Barber et al., 2023; Lei et al., 2018).

Exchangeability is only approximately satisfied. Monthly `tas` fields exhibit temporal autocorrelation through seasonality and low-frequency modes such as ENSO. The LOO construction ensures that the calibrated ESM is excluded from the corresponding teacher’s target supervision, but months within an ESM are not strictly i.i.d. draws from a stationary distribution. Adaptive and sequential conformal methods (Gibbs and Candès, 2021; Xu and Xie, 2023; Lee et al., 2024) provide a more principled route for time-dependent calibration. In

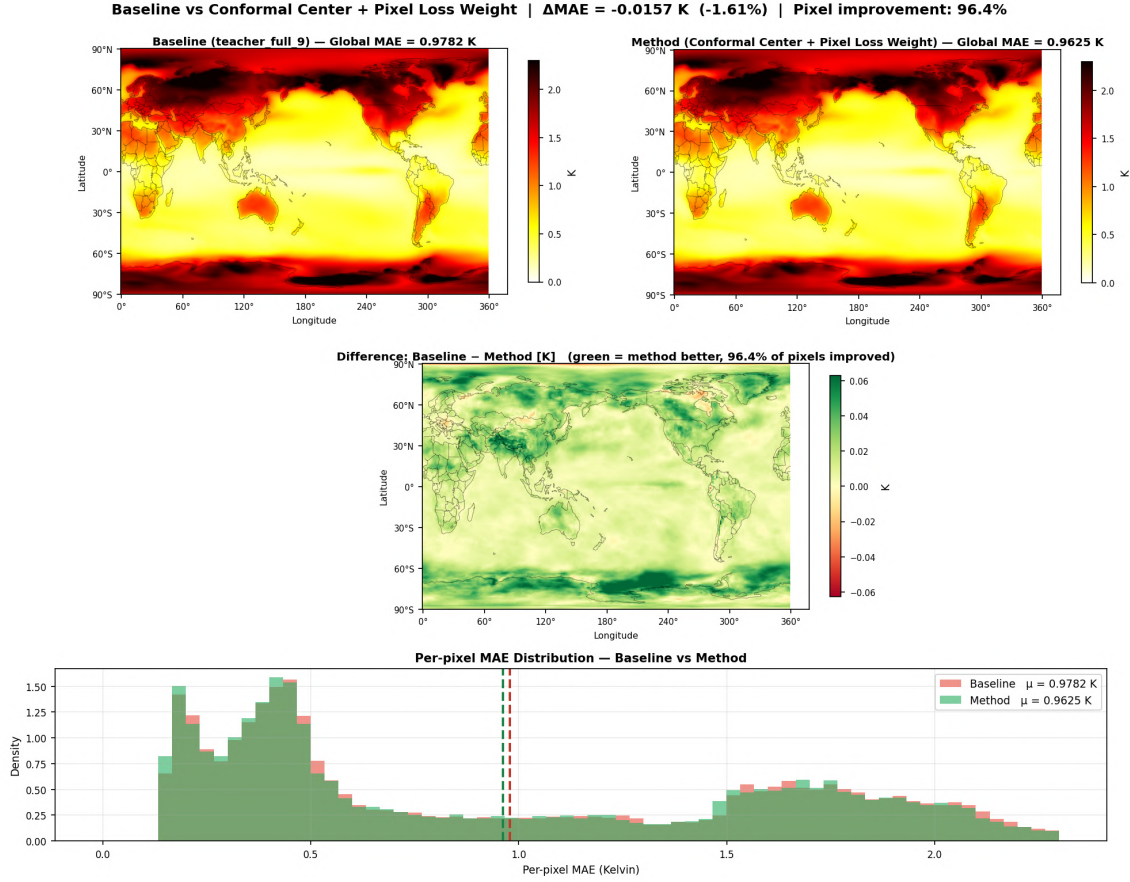


Figure 6: Spatial MAE comparison. The figure shows baseline MAE, student MAE, the difference map (baseline minus student; positive values indicate improvement), and the distribution of per-pixel MAE values. The student reduces error across 96.4% of grid cells.

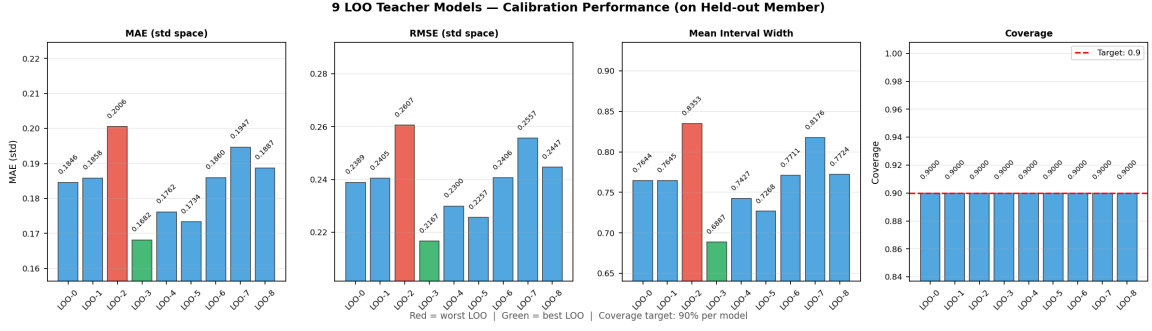


Figure 7: LOO teacher calibration summary. The panels show calibration-set MAE, RMSE, mean conformal half-width, and empirical coverage for each teacher on its held-out target ESM. The conformal threshold (Equation (2)) targets nominal 90% marginal coverage under exchangeability.

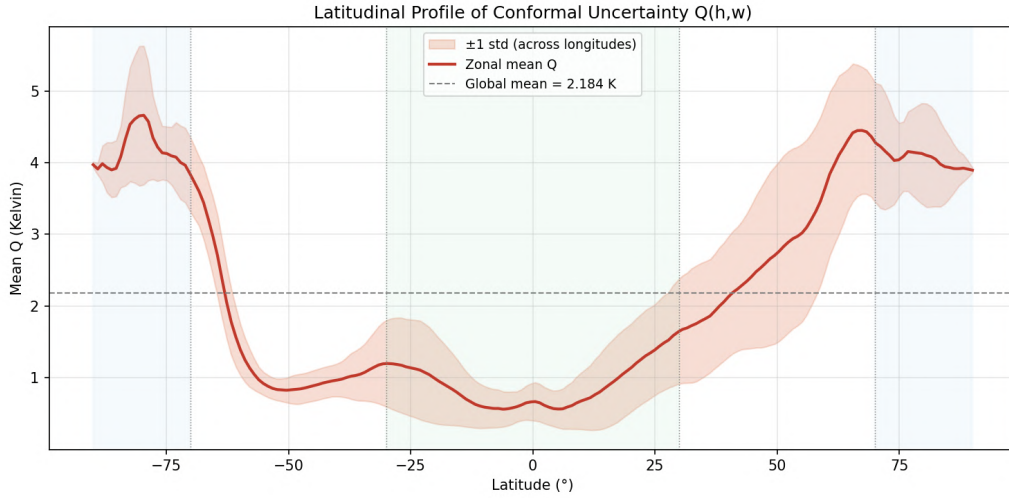


Figure 8: Latitudinal profile of the mean conformal half-width $Q(h,w)$ in Kelvin (zonal mean ± 1 std across longitudes). Blue shading marks Arctic and Antarctic bands; green marks the Tropics. Polar regions require substantially wider conformal intervals than the tropics, reflecting the higher difficulty of emulating high-latitude climate variability.

the present work we use standard split CP because it gives a transparent per-pixel marginal baseline and because the student coverage is reported only as an empirical calibration check.

Student aggregation weakens the formal guarantee. The distilled student is trained using post-calibration teacher statistics. The interval in Equation (13) is centred at the student prediction but uses a teacher-calibration-derived half-width. Therefore the reported 90.6% coverage is an empirical calibration check, not a theorem about g . Calibrating the final student directly on a separate held-out calibration set would restore a formal split-CP guarantee for the deployed predictor, at the cost of an additional data split.

Single variable, single dataset. We study near-surface air temperature on a single nine-member CMIP6 subset. Temperature residuals are relatively well suited to symmetric absolute-residual intervals. Variables with heavier-tailed, skewed, or zero-inflated residuals, such as precipitation and wind extremes, may require asymmetric intervals or quantile-regression-based nonconformity scores. Conformalized quantile regression (Romano et al., 2019) is the most natural extension in that direction. Generalisation across CMIP families, reanalysis products, and additional climate variables is left to future work.

Limited ablation and significance analysis. The global MAE improvement of 1.61% is modest in absolute terms. The regional gains and 96.4% pixel-improvement rate suggest that the benefit is spatially widespread, but paired significance tests and ablations separating center weighting, pixel-loss weighting, and teacher distillation would further strengthen the empirical claim.

8. Conclusion

We introduced a calibration-guided distillation framework for global temperature emulation that uses split conformal calibration not only as a post-hoc uncertainty wrapper, but also as a source of spatial training signal. By training leave-one-out teacher emulators, calibrating each teacher on its excluded CMIP6 member, and converting the resulting pixelwise residual statistics into teacher-consensus and pixel-loss weights, the proposed method links uncertainty calibration with ensemble compression in a single student model.

On monthly near-surface air temperature emulation, the distilled student improves over a strong UNet++ baseline, reducing global MAE from 0.9782 K to 0.9625 K and improving 96.4% of grid cells. The largest absolute gains occur in polar regions, where calibration residuals and conformal half-widths indicate higher emulator uncertainty. The resulting student intervals achieve 90.6% empirical pixelwise coverage, which we interpret as an empirical calibration check rather than a formal conformal guarantee for the distilled predictor.

The main message of this work is that conformal residuals can provide useful spatial information beyond interval construction: they can identify where an emulator is unreliable and guide training toward those regions. Future work will calibrate the final student on an independent calibration split to recover a formal split-conformal guarantee for the deployed model, evaluate sequential conformal variants for temporally dependent climate fields, and extend the framework to heavier-tailed climate variables such as precipitation and wind extremes.

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References

- Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.
- Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845, 2023.
- Veronika Eyring, Sandrine Bony, Gerald A. Meehl, Catherine A. Senior, Bjorn Stevens, Ronald J. Stouffer, and Karl E. Taylor. Overview of the coupled model intercomparison project phase 6 (CMIP6) experimental design and organization. *Geoscientific Model Development*, 9(5):1937–1958, 2016.
- Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems*, volume 34, pages 1660–1672, 2021.
- Tilmann Gneiting and Matthias Katzfuss. Probabilistic forecasting. *Annual Review of Statistics and Its Application*, 1:125–151, 2014.
- Vignesh Gopakumar, Joel Oskarsson, Ander Gray, Lorenzo Zanisi, Stanislas Pamela, Daniel Giles, Matt Kusner, and Marc Deisenroth. Valid error bars for neural weather models using conformal prediction. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*. PMLR, 2024.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Junghwan Lee, Chen Xu, and Yao Xie. Transformer conformal prediction for time series, 2024.
- Jing Lei, Max G’Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018.
- Henrik Linusson, Ulf Johansson, and Henrik Boström. Efficient conformal predictor ensembles. *Neurocomputing*, 397:266–278, 2020. doi: 10.1016/j.neucom.2019.07.113.
- Andrey Malinin, Bruno Mlodozieniec, and Mark Gales. Ensemble distribution distillation. In *International Conference on Learning Representations*, 2020.

- Fan Meng and Tao Song. Uncertainty forecasting system for tropical cyclone tracks based on conformal prediction. *Expert Systems with Applications*, 249:123743, 2024. doi: 10.1016/j.eswa.2024.123743.
- Harris Papadopoulos, Kostas Proedrou, Vladimir Vovk, and Alexander Gammerman. Inductive confidence machines for regression. In *European Conference on Machine Learning*, pages 345–356. Springer, 2002.
- Stephan Rasp and Nils Thuerey. Data-driven medium-range weather prediction with a Resnet pretrained on climate simulations. *Journal of Advances in Modeling Earth Systems*, 13(2), 2021.
- David Rolnick, Priya L. Donti, Lynn H. Kaack, Kelly Kochanski, Alexandre Lacoste, Kris Sankaran, Andrew Slavin Ross, Nikola Milojevic-Dupont, Natasha Jaques, Anna Waldman-Brown, et al. Tackling climate change with machine learning. *ACM Computing Surveys*, 55(2), 2022. doi: 10.1145/3485128.
- Yaniv Romano, Evan Patterson, and Emmanuel Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Tapio Schneider, Shiwei Lan, Andrew Stuart, and João Teixeira. Earth system modeling 2.0: A blueprint for models that learn from observations and targeted high-resolution simulations. *Geophysical Research Letters*, 44(24):12396–12417, 2017. doi: 10.1002/2017GL076101.
- Alper Unal, Busra Asan, Ismail Sezen, Bugra Yesilkaynak, Yusuf Aydin, Mehmet Ilicak, and Gozde Unal. Climate model-driven seasonal forecasting approach with deep learning. *Environmental Data Science*, 2:e29, 2023.
- Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- Chen Xu and Yao Xie. Sequential predictive conformal inference for time series. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 38707–38727. PMLR, 2023.
- Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. Unet++: A nested u-net architecture for medical image segmentation. In *International workshop on deep learning in medical image analysis*, pages 3–11. Springer, 2018.

Appendix A. Notation Summary

Symbol	Definition
α	Significance level (= 0.10)
q_{level}	Finite-sample-corrected quantile level
$s_i(t, h, w)$	Nonconformity score at pixel (h, w) and time t for teacher i

$Q_i(h, w)$	Per-pixel conformal half-width for LOO teacher i
$\bar{Q}(h, w)$	Mean teacher-calibration-derived half-width used for empirical student intervals
$\delta_i(h, w)$	Mean calibration bias / centre shift of teacher i at pixel (h, w)
$\tilde{f}_i$	Bias-corrected teacher i prediction
$\bar{f}$	Bias-corrected ensemble consensus
w_i^{dyn}	Dynamic teacher weight (time-varying softmax)
w_i^{static}	Static diagnostic teacher weight (calibration-time mean)
β	Softmax temperature for center weighting (= 2.0)
$p(h, w)$	Pixel-loss weight (normalised mean teacher MAE)
λ	Distillation loss coefficient (= 1.0)
